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ROTATION POWERED PULSARS IN THE X-RAYS

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by
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Abstract

The dissertation focuses on the study of rotation-powered pulsars, the primary observational manifestation of neutron stars. These objects are powerful sources of electromagnetic radiation and relativistic particles whose emission is provided by the loss of pulsar rotational energy. Understanding the evolution of pulsars, which happens over billion year timescales, requires detection and study of pulsars at different stages of evolution. I present detailed X-ray analyses of pulsars at four distinct stages of evolution and compare their emission behavior with that of other pulsars expected to be in similar evolutionary stages. I also show key characteristics of the pulsars that make them unique in their group.

I start with a young and energetic pulsar, PSR J2022+3842 (characteristic age $\tau_c \approx 9$ kyr, spin-down power $\dot{E} = 3 \times 10^{37}$ erg s^{-1}), with powerful non-thermal emission. X-ray timing of the pulsar revealed double-peaked X-ray profile with a period twice the previously established value. Our analysis allowed us to update the pulsar's spin-down power and X-ray efficiency using the correct timing results, which brought the pulsar more in-line with other young X-ray pulsars. I also provide the phase-dependent behavior of the pulsar's non-thermal emission. Pulsars with true ages, often substituted by characteristic age, below $\tau_c \sim 100$ kyr are considered young and ones with $\tau_c \gtrsim 1$ Myr are considered old, with the 'middle-aged' pulsars in the middle. My next pulsar is a $\tau_c = 1.8$ Myr old J1836+5925 ($\dot{E} = 1 \times 10^{34}$ erg s^{-1}), which is perhaps the brightest X-ray source among the oldest pulsars still observable in the γ-rays. Detailed timing and spectral analyses show strong evidence of an absorption feature (perhaps an electron cyclotron line) in the pulsar's spectrum. Characterizing its thermal emission might have important implications for the neutron star cooling models. Moving another two orders of magnitude up in τ_c, we arrive at one of the oldest known non-recycled X-ray pulsars, PSR J0108-3430, with $\tau_c = 166$ Myr and $\dot{E} = 5.8 \times 10^{30}$ erg s^{-1}. The pulsar's spectrum likely consists of a thermal component, emitted from a hot polar cap, and a non-thermal component, emitted from its magnetosphere. The X-ray pulse profile shows a single, asymmetric peak which could be explained by an axially-asymmetric

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temperature distribution at the pole or by the non-thermal emission from the outer gap. The three pulsars represent important stages in the evolutionary path that a hypothetical single young pulsar like J2022+3842 might take, as it passes through stages close to γ-ray emission turn-off (like J1836+5925) and X-ray turn-off (similar to J0108-3430).

Pulsars in binaries can follow an alternative path. By accreting matter from their companions they can be 'recycled' to short millisecond periods and emit X-rays and γ-rays for billions of years. I also present a special class of such recycled pulsars which are believed to be in the process of fatally ablating their companions. I present the X-ray analysis of PSR J1446-4701, an $\dot{E} = 3.6 \times 10^{34}$ erg s^{-1} pulsar in a 6.7 hr binary orbit, and PSR J1311-3430, an $\dot{E} = 4.9 \times 10^{34}$ erg s^{-1} pulsar in an extreme 1.6 hr binary orbit. PSR J1446-4701 turned out to be a non-eclipser with possibly low (face-on) orbital inclination, with emission from both the pulsar and the intra-binary shock observable throughout the binary orbit. PSR J1311-3430 is a known eclipser, in which hints of spectral variability have been found, between pulsar superior and inferior conjunction phases. I also present a comprehensive comparison of the sample of such extremely low-mass binary pulsars. We reveal the true nature of pulsars, slowly and steadily, usually one target at a time, but eventually we expect useful patterns to emerge that improves our understanding of the population of rotation powered pulsars.

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Chapter 1
Introduction

Neutron stars are the collapsed cores of massive stars supported against gravitational collapse through neutron degeneracy pressure. Their existence was originally proposed by Walter Baade and Fritz Zwicky (Baade & Zwicky 1934) soon after the discovery of neutrons, but the possibility of detecting emission from such an object was not entertained until 1967 (Pacini 1967). The first neutron star was discovered through detection of periodic radio pulses by Jocelyn Bell and Anthony Hewish in 1967. The uniquely periodic nature of radio pulses from neutrons stars have earned them the nickname pulsars or radio pulsars. A majority of the pulsars have been detected at radio frequencies. There have been close to 2500 pulsars detected so far, almost all of which have been detected in the radio, with the exception of a handful of radio-quiet pulsars, which have been detected only at X-ray and/or γ-ray energies.

During the core collapse of the progenitor star to form a neutron star, the stellar angular momentum and magnetic flux are conserved and 'concentrated' as the core is scaled down orders of magnitude in size. The newly formed neutron stars can reach spin periods of a few milliseconds and the magnetic energy density is concentrated, producing surface magnetic fields typically of the order of 10^{12} G. Meanwhile, the collapse causes the core temperatures to reach $\sim 10^{11}$ K, and densities $\rho \approx 4 \times 10^{14}$ g cm^{-3} exceeding the nuclear density ($\rho_{\rm nuc} = 2.3 \times 10^{14}$) (Kaspi et al. 2006). These pulsars are primarily powered by the energy loss due to rotational spin-down, hence they are called rotation-powered pulsars (RPP).

1.1 Neutron Star Classifications

RPPs make up the majority of the neutron stars detected, but recent observations, particularly in X-rays and γ-rays, have shown a great diversity of neutron star types. Neutron stars can be broadly classified, according to the primary source of energy that accounts for their electromagnetic and particle emission, into the following types.

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1. Rotation-powered pulsars: The rotational energy loss of the pulsar powers the radiation via creation and acceleration of e^+e^- pairs in a strong magnetic field.
2. Thermally-powered pulsars: The cooling neutron stars are powered by the thermal emission from their surface observable in soft X-rays.
3. Magnetically-powered pulsars: The energy of very high magnetic fields is released through crustal heating and cracking, leading to powerful bursts and 'quiescent' X-ray emission.
4. Accretion-powered pulsars: A binary companion transfers material onto the spinning neutron star, and the gravitational potential energy is responsible for bulk of the observed emission.

In this dissertation, I focus on rotation-powered pulsars in their non-recycled evolutionary stages in isolation, and also on a special class of interacting, recycled pulsars in binaries.

1.2 Pulsar Emission

The most important empirical properties of a pulsar are its period P and period derivative $\dot{P}$, from which a number of pulsar properties can be derived, assuming it loses the rotational (spin) energy as a rotating magnetic dipole. A rotating dipole emits electromagnetic radiation and the back reaction torque reduces the angular momentum of the rotator. The strength of the torque depends on the inclination between the rotation and the magnetic axis, with an orthogonal rotator experiencing the maximum torque. The strength of the torque is in general proportional to Ω^n, where $\Omega = 2\,\pi/P$ is the angular velocity and n is the braking index. This simple model can be applied to pulsars, which allows one to calculate a set of simple parameters.

1. Spin-down power ($\dot{E}$): Since the rotational energy of a pulsar is $E = I\,\Omega^2/2$, where I is the moment of inertia, the spin-down power can be calculated using the expression $\dot{E} = -4\,\pi^2\,\dot{P}/P^3$. For an assumed mass $M_{\rm NS} = 1.4$ and radius $R_{\rm NS} = 10$ km, $I \approx 10^{45}$ g cm^2.
2. The characteristic age ($\tau_{\rm c}$) is calculated for a magnetic braking index of n and an initial spin period of P_0 using the expression

$$\tau_{\rm c} = \frac{P}{(n-1)\dot{P}}\left[1 - \left(\frac{P_0}{P}\right)^{n-1}\right] \tag{1.1}$$

   It may be simplified to $\tau_{\rm c} \approx P/2\dot{P}$ s, for pulsars that have appreciably slowed down from a short initial period. It is important to note that the characteristic age of a pulsar may differ from the true age.

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3. The dipole magnetic field strength $B = \sqrt{\left(3\,c^3 I P \dot{P}/8\,\pi^2\,R_{\rm NS}^6\right)}$, where c is the speed of light. Note that the actual magnitude of the surface magnetic field can differ from this value by a factor of a few.

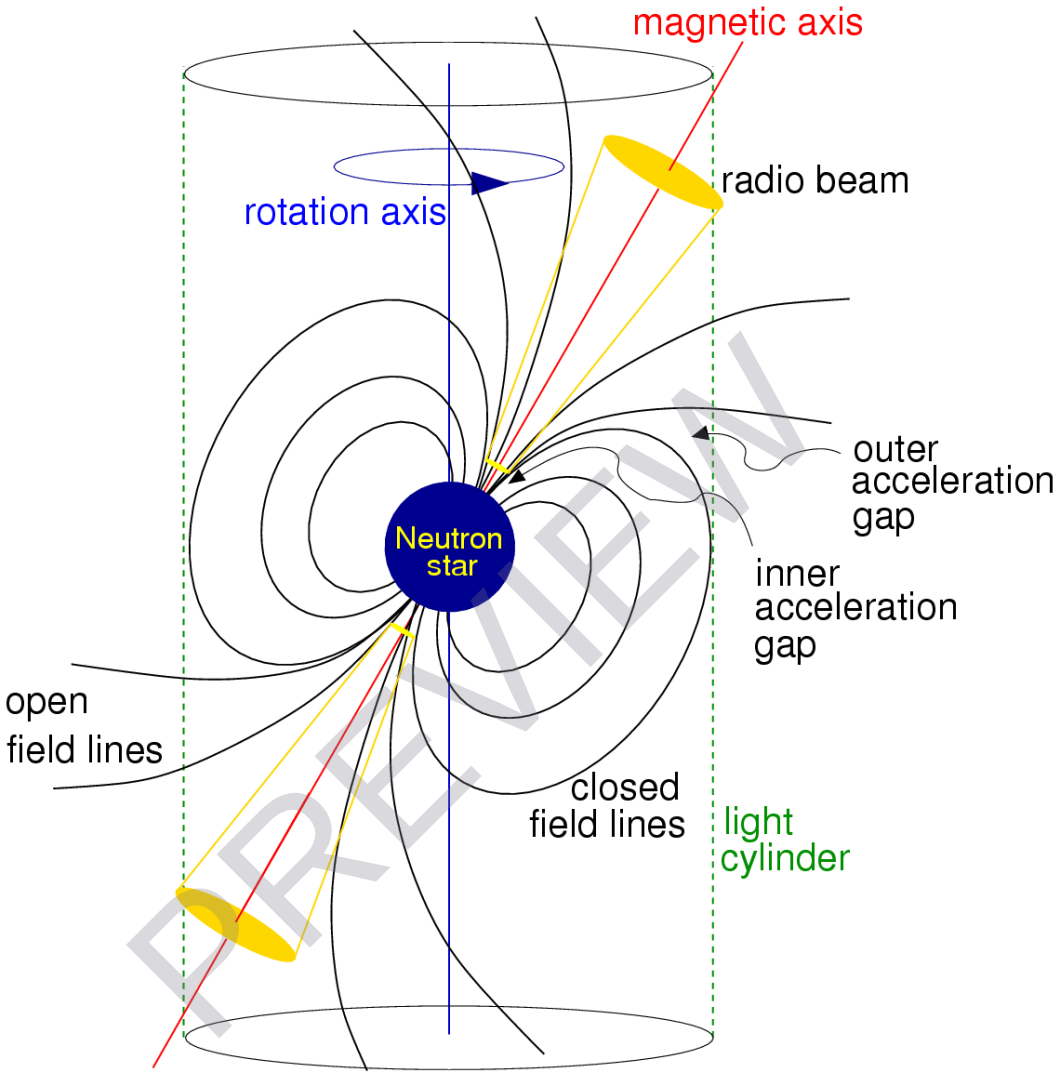
The rotating dipole model is a gross simplification focused simply on a dipole magnetic field losing its rotational energy in vacuum. Even though the true nature of the pulsar's emission mechanism is not fully understood, improved models have been developed from theoretical considerations and observational evidence. Approximating the neutron star as a uniformly magnetized sphere produces Lorentzian force-free charge configurations in its interiors (Goldreich & Julian 1969). The surface charge distribution, however, induces a quadrupole field producing electric forces 10 orders of magnitude greater than the gravitational forces, which allows extraction of charged partcles from the surface to populate the magnetosphere. These charged particles undergo charge separation and assume a global force-free configuration in the magnetophere. The plasma-filled magnetosphere co-rotates with the neutron star up to the light cylinder radius, beyond which the co-rotation speeds exceed the speed of light, and loses charges through the open field lines that extend beyond.

Models that explain the non-thermal emission observed from pulsars require presence of gaps in the charge distribution, where particles can be accelerated to ultra-relativistic energies. Creation of such gaps could occur along open field lines when the particles lost by the wind are not replenished (Holloway 1975). Models that utilize such gaps near the magnetic pole (polar gap: Harding & Muslimov 1998), near the last open field line extending from the polar gaps up to the light cylinder (slot gap: Muslimov & Harding 2003), and above the null surface which forms the boundary of the charge separation (outer gap: Romani 1996, Cheng et al. 2000), have been successful in explaining the multiwavelength emission of pulsars. Figure 1.1, adopted from Lorimer & Kramer (2012), illustrates a pulsar and its magnetosphere depicting the rotation and magnetic axes, light cylinder, open and closed field lines, and the approximate location of the gaps.

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1.2.1 Radio Emission

There are no generally accepted models to explain the highly coherent and polarized radio emission observed in pulsars (Lorimer & Kramer 2012). Curvature radiation from bulk acceleration and movement of charges along magnetic field lines, proposed in early years (Komesaroff 1970; Ruderman & Sutherland 1975) can explain the coherent emission, but lacks a reasonable physical process that could produce such confinements and bulk motions of particles. Free electron maser mechanism (Melrose 1989; Rowe 1995) have some favorable qualities, but it is still far from being established as the true model for explaining the radio emission. Significant progress has been made in the field of radio pulse profile modeling, which focuses on the



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Figure 1.1: A pulsar magnetosphere toy model adopted from Lorimer & Kramer (2012).

origin and distribution of particles that produce the pulsed radio emission (Rankin 1983; Lyne & Graham-Smith 2006 and references therein). Radio emission from pulsars are expected to originate from near the magnetic pole, in gaps where charge acceleration leads to pair-cascades. The particles channeled along the narrow paths along the open field lines are believed to produce the stable radio beams with fairly narrow pulse profiles. Models that describe a cone of radio emission around the magnetic axis (Radhakrishnan & Cooke 1969; Komesaroff 1970; Rankin 1993) are